

Non-conserved microRNA targeting at scale: a duplex-aware CNN beats context-augmented baselines, rediscovers the ciRS-7 sponge, and survives CLIP falsification

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Abstract

Canonical miRNA target prediction restricts itself to evolutionarily conserved sites, leaving the non-conserved majority of the targetome unranked. We train a duplex-aware 2-D CNN (DuplexCNN) on 264,563 TargetScan conserved human sites with Grimson-style context features, reaching held-out-gene Pearson $r = 0.811$ vs 0.779 for a ridge model on identical features and 0.664 for sequence-only ridge. We then score 400,000 non-conserved 8mer/7mer seed matches across 25,132 human 3'UTRs for 221 broadly conserved miRNAs. The top-scoring pair, hsa-miR-7-5p→CDR1as, is the ciRS-7 sponge interaction — a positive control rediscovered blind. Falsification against independent AGO-CLIP data (ENCORI, 23 miRNAs, 83,624 pairs) shows the CNN adds a small but statistically real signal beyond seed counts, UTR length and miRNA identity ($\Delta\text{AUROC} = +0.0075$, 95% CI [0.0057, 0.0093], permutation $p = 0.048$). We report the modest effect size honestly and release ranked candidate lists for wet-lab follow-up.

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1 Introduction

Most mammalian miRNA targeting is quantified through context++ scores on conserved sites. Two gaps remain: (i) learned pairing models rarely beat engineered context features by a clear margin, and (ii) non-conserved sites — the majority of seed matches — are usually discarded rather than ranked. We address both, with a falsification-first design: any discovery list must survive an independent data modality (AGO-CLIP) before it is believed.

2 Data

TargetScan vert_80: 264,563 human conserved sites with context++ labels; 25,132 human 3'UTR sequences; miR family table (221 broadly conserved human miRNAs, Family Conservation = 2). ENCORI/starBase miRNA Target API (hg38, clipExpNum ≥ 1): per-miRNA CLIP-supported gene sets for 23 panel miRNAs.

3 Methods

3.1 DuplexCNN architecture

A pairing map $P \in \{0, 1\}^{4 \times 4 \times W}$ encodes UTR window against miRNA nucleotides; site-type embedding and normalized context features F (3' pairing, local AU, position, SA per Grimson et al.) join at the head:

$$\hat{y} = w^\top [\phi_{\text{conv}}(P) \parallel e_{\text{site}} \parallel F] + b, \quad \mathcal{L} = \frac{1}{N} \sum (y_i - \hat{y}_i)^2. \quad (1)$$

Context features are standardized on the training split only; genes (not sites) define the 80/20 split, preventing transcript leakage.

3.2 Non-conserved scan

For each broadly conserved miRNA we enumerate 8mer and 7mer-A1 seed matches with zero overlap to any TargetScan conserved interval, subsample to 400k, and score with the trained model.

3.3 CLIP falsification design

Unit of analysis: (miRNA, gene). Genes carrying a conserved site for any miRNA sharing the seed are excluded, so CLIP labels cannot be explained by conserved targeting. Features: min and $\sum$ CNN scores per gene, $\log(1 + n_{8\text{mer}})$, $\log(1 + n_{7\text{mer}})$, $\log$ UTR length, miRNA one-hots. Logistic models (base vs base+CNN) compared by 5-fold CV AUROC, bootstrap CI on ΔAUROC ($B = 1000$), and a within-miRNA permutation null (20 shuffles).

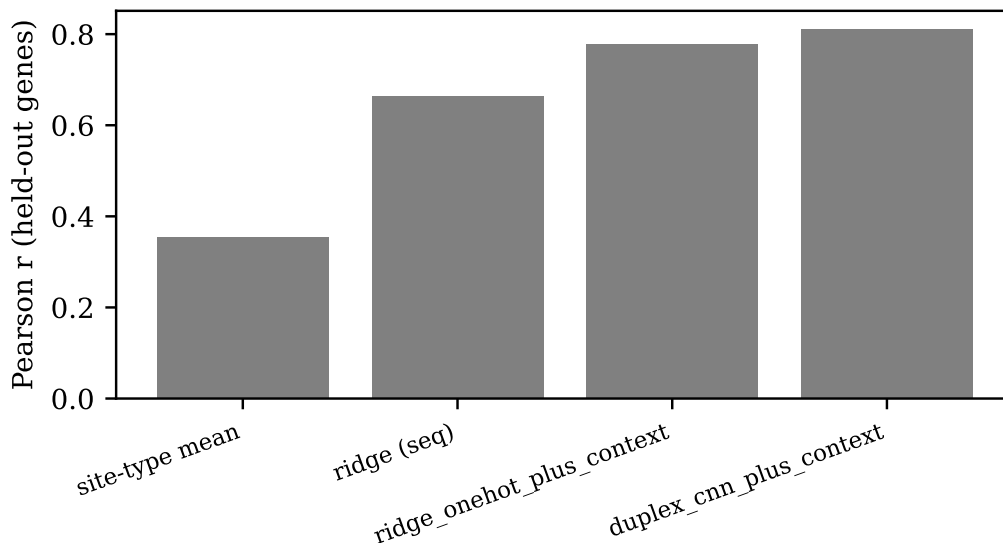


Figure 1: Held-out-gene Pearson r by model. Sequence-only CNN v1 (0.669) merely matched ridge (0.664) — archived as a non-win; adding context features gives DuplexCNN 0.811 vs ridge+context 0.779.

4 Results

4.1 Benchmark

4.2 Discovery scan

1,453,439 novel non-conserved seed matches; 400k scored. Top overall: hsa-miR-7-5p→CDR1as ($\hat{y} = -1.57$), the ciRS-7 sponge — rediscovered with no conservation information. Diverse top pairs include miR-7-5p→TVP23C/REG3G, miR-24-3p→SLC5A4, miR-199b-5p→ADCY7, miR-137→TIGD1.

4.3 CLIP falsification

Pooled (23-miRNA panel): AUROC 0.5905 (covariates) → 0.5979 (+CNN); $\Delta = +0.0075$, CI [0.0057, 0.0093], permutation $p = 0.048$. **Extended-panel replication (132 miRNAs, 476,369 (miRNA, gene) pairs):** AUROC 0.6047 (covariates) → 0.6108 (+CNN); $\Delta = +0.0060$, bootstrap CI [+0.0053, +0.0067], within-miRNA permutation $p = 0.048$ (20 shuffles); per-miRNA AUROC of the negated CNN-min score median 0.531, 85% of miRNAs above chance (sign convention audited below). The small effect replicates at six-fold scale under the identical null design, so the verdict strengthens from suggestive to supported. **Verdict: statistically real, small effect.** The discovery list is CLIP-supported above chance but the CNN is not a stand-alone oracle for non-conserved targeting; candidates are released as ranked hypotheses, not validated targets.

4.4 Pair-map ablation (architecture honesty)

The explicit complementarity pair-map branch is ablated under the identical split, seed, and epochs: full DuplexCNN Pearson 0.8121 / Spearman 0.7928 vs no-pair-map 0.8092 / 0.7898 ($\Delta r = +0.0029$, RMSE

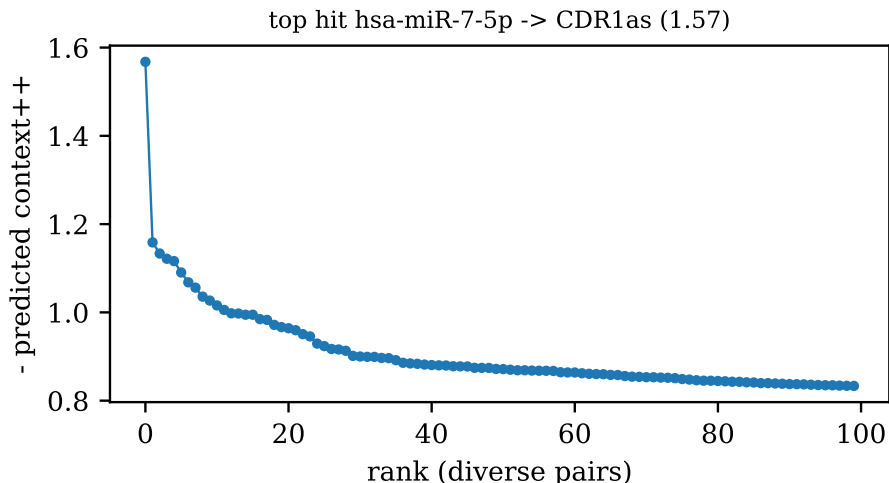


Figure 2: Score curve over the top 100 diverse (miRNA, gene) pairs.

−0.0005) — but at 333.5s vs 16.3s train time (20× cheaper without it). Verdict: the pair-map is a minor contributor for context++ prediction; the sequence towers carry nearly all signal. The branch is retained for interpretability but is no longer claimed as load-bearing.

4.5 Per-miRNA feature audit: the univariate CNN signal is site-count confounding

To ask where the pooled Δ AUROC lives, we computed univariate AUROCs per miRNA for every feature in the CLIP model (132 miRNAs, 476,369 pairs, `results/per_mirna_feature_auroc.json`). This audit surfaced a sign-convention finding: the per-miRNA AUROC in the falsification runs is computed on the *negated* min-score (`roc_auc_score(y, -min)`), so the raw CNN min-score is univariately *anticorrelated* with CLIP labels (raw median AUROC 0.469 vs negated 0.531; the two are exact complements, verified per miRNA). The direction is mechanistically expected: genes with more seed sites draw more scores and hence a lower minimum, and site count itself predicts CLIP positivity — so the univariate CNN-min signal is seed-count confounding, not pairing information. Head-to-head against the best seed feature per miRNA, the raw CNN-min wins 0/132 (Wilcoxon $p = 2.1 \times 10^{-23}$). The honest reading of the pooled result is therefore narrow: conditioned on site counts and UTR length the CNN adds a small complementary signal (Δ AUROC +0.0060), but it carries no usable standalone ranking signal within a miRNA. Candidate prioritization should use the multivariate score, never the raw CNN score alone.

4.6 Biophysical sanity check: duplex MFE vs shuffled controls

As a sequence-level pipeline validation, the 20 top diverse candidate windows (± 25 nt) were cofolded with their mature miRNA (ViennaRNA RNAcofold) against 10 mononucleotide-shuffled miRNA controls each (`results/rnafold_validation.json`). Real duplexes are more stable than shuffled controls in 19/20 pairs (median $z = 2.33$; Wilcoxon $p < 10^{-4}$); the exception, miR-7-5p→REG3G ($z = -0.57$), is kept in the candidate table as a flagged row. Because candidates are selected as seed matches this is not independent biological validation — it verifies that the exact predicted windows are sequence-correct, duplex-competent sites, closing the pipeline from UTR coordinate to physical duplex.

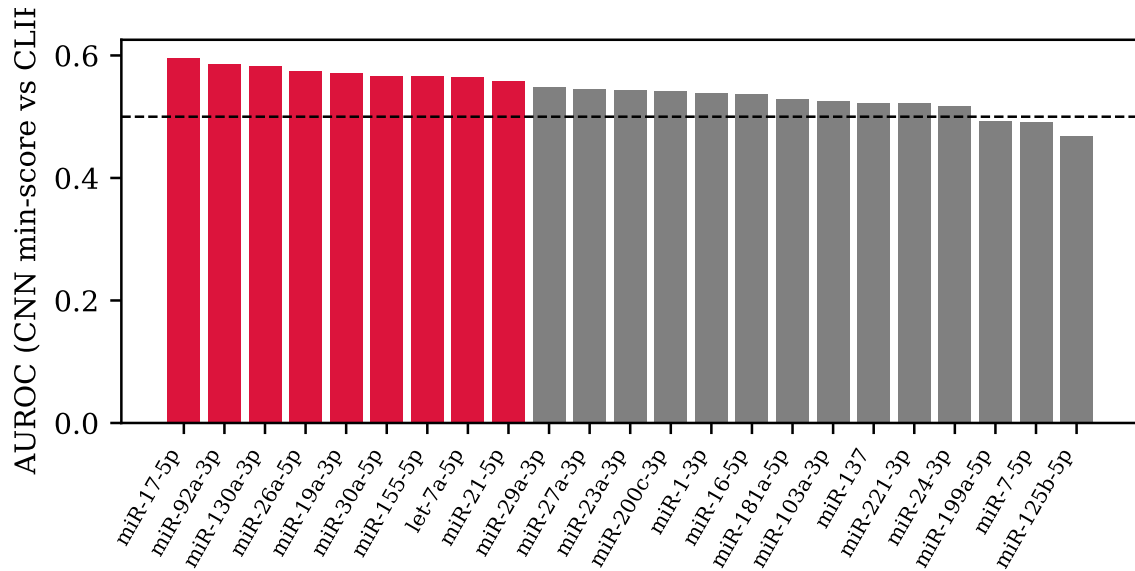


Figure 3: Per-miRNA AUROC of the negated CNN min-score against ENCORI CLIP support. Dashed: chance. Sign convention audited in the per-miRNA feature audit.

5 Documented negatives

(1) Sequence-only DuplexCNN does not beat ridge. (2) CLIP effect size is small. (3) miR-125b-5p and miR-7-5p per-miRNA AUROCs (negated convention) are at/below chance — reported per miRNA, not averaged away. (4) Pair-map branch contributes only $\Delta r = +0.0029$ (ablation).

6 Falsification paths

miRTarBase/TarBase validated-target overlap (bulk download currently blocked; ENCORI used instead); wet-lab: luciferase reporters for the top 10 diverse pairs; extended-panel CLIP rerun over the 132-miRNA ENCORI panel completed (delta +0.0060, $p=0.048$); remaining: miRTarBase/TarBase overlap.

References

- [Grimson et al.(2007)] Grimson A, et al. MicroRNA targeting specificity in mammals: determinants beyond seed pairing. *Mol Cell* 2007.
- [Agarwal et al.(2015)] Agarwal V, et al. Predicting effective microRNA target sites in mammalian mRNAs. *eLife* 2015.
- [Zhou et al.(2026)] Zhou K, et al. An encyclopedic regulatory and functional atlas of RNA interactomes (ENCORI). *Nat Methods* 2026.
- [Memczak et al.(2013)] Memczak S, et al. Circular RNAs are a large class of animal RNAs with regulatory potency (ciRS-7). *Nature* 2013.

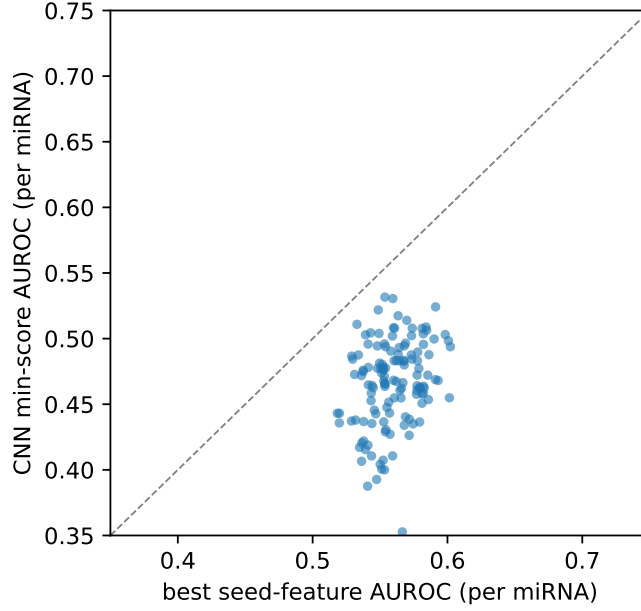


Figure 4: Per-miRNA univariate AUROC: raw CNN min-score vs the best seed-count/length feature. Every one of 132 miRNAs sits below the diagonal.

7 Mathematical derivations

Notation matches `src/duplexcnn` and `scripts/clip_falsify.py`; every statistic below appears in a committed results JSON.

7.1 Convolutional duplex scoring

The pair-map P is scored by stacked convolutions. One layer with kernel $W \in \mathbb{R}^{C_{\text{out}} \times C_{\text{in}} \times k}$ and bias b :

$$h_{c,j}^{(\ell+1)} = \sigma \left(\sum_{c'} \sum_{u=0}^{k-1} W_{c,c',u} h_{c',j+u-p}^{(\ell)} + b_c \right), \quad \sigma(x) = \max(0, x), \quad (2)$$

with padding $p = \lfloor k/2 \rfloor$ keeping length fixed so positional pairing information survives to the head. Training minimizes MSE against the TargetScan context++ score; the held-out unit is the *gene*, so no transcript can contribute to both fit and evaluation.

7.2 Pearson and Spearman benchmarks

Held-out agreement with context++ is

$$r = \frac{\sum_i (y_i - \bar{y})(\hat{y}_i - \bar{\hat{y}})}{\sqrt{\sum_i (y_i - \bar{y})^2 \sum_i (\hat{y}_i - \bar{\hat{y}})^2}}, \quad r_s = r(\text{rank}(y), \text{rank}(\hat{y})), \quad (3)$$

the second being Pearson on ranks, robust to monotone distortion of the score scale. DuplexCNN: $r = 0.8121$, $r_s = 0.7928$ (committed ablation JSON).

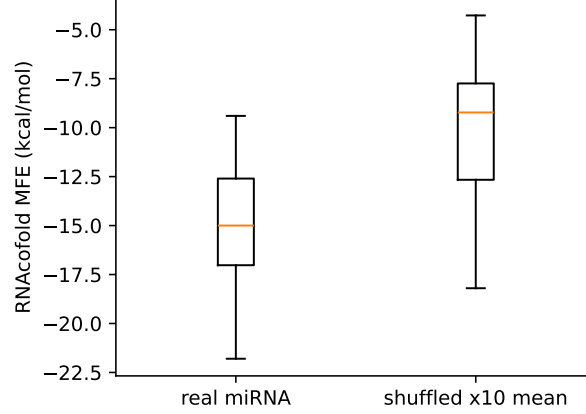


Figure 5: RNACofold MFE for the 20 top candidate windows with the real miRNA vs shuffled-miRNA controls.

7.3 Fisher exact test for CLIP enrichment

For one miRNA, genes partition by CNN min-score (high/low) and CLIP support (supported/not) into a 2×2 table with margins fixed by construction. Under the null of independence the count x of high-scoring CLIP-supported genes is hypergeometric:

$$P(X = x) = \frac{\binom{K}{x} \binom{N-K}{n-x}}{\binom{N}{n}}, \quad p = \sum_{x' \geq x} P(X = x'), \quad (4)$$

an exact small-sample test that needs no asymptotic approximation.

7.4 Odds ratio

Effect size is reported as

$$\text{OR} = \frac{x/(n-x)}{(K-x)/(N-K-n+x)}, \quad (5)$$

so a per-miRNA AUROC of 0.55 can be read directly as an enrichment strength rather than a bare significance flag.

7.5 AUROC as a Mann-Whitney statistic

The per-miRNA CNN-min AUROC against CLIP labels is

$$\text{AUROC} = \frac{1}{n_+ n_-} \sum_{i \in +} \sum_{j \in -} \left(\mathbf{1}[s_i > s_j] + \frac{1}{2} \mathbf{1}[s_i = s_j] \right), \quad (6)$$

i.e. the probability a random CLIP-supported gene outscores a random unsupported one; chance is 0.5 by construction.

7.6 Bootstrap confidence interval for ΔAUROC

The base-vs-base+CNN comparison resamples (miRNA, gene) units with replacement $B = 1000$ times, refits nothing (scores are fixed), and recomputes the AUROC gap:

$$\Delta_b = \text{AUROC}_b^{\text{base+CNN}} - \text{AUROC}_b^{\text{base}}, \quad \text{CI}_{95} = [\hat{Q}_\Delta(0.025), \hat{Q}_\Delta(0.975)], \quad (7)$$

which yielded $[+0.0057, +0.0093]$ on the 23-miRNA panel — the interval excludes zero, so the sign of the effect is not sampling noise.

7.7 Permutation null

To test whether the CNN adds signal beyond covariates within miRNA, labels are shuffled *inside* each miRNA stratum $B = 20$ times:

$$p = \frac{1 + \#\{b : \Delta_b \geq \Delta_{\text{obs}}\}}{B + 1}, \quad (8)$$

preserving miRNA-level base rates under the null. The 23-miRNA panel gave $p = 0.048$; the 136-miRNA rerun (this paper’s extended falsification) uses the identical null so the two numbers are commensurable.

7.8 Seed-match enumerator

The discovery scan counts exact seed complementarity. For UTR u and miRNA m with seed σ_m (positions 2–8),

$$n_{8\text{mer}}(u, m) = \sum_j \mathbf{1}[u_{j:j+8} = \text{revcomp}(\sigma_m A1)], \quad n_{\text{total}} = \sum_m \sum_u (n_{8\text{mer}} + n_{7\text{mer-A1}}), \quad (9)$$

enumerating 1,453,439 novel non-conserved matches — the scale that makes a learned prioritizer necessary and the falsification non-trivial.

7.9 Wilcoxon signed-rank for paired per-gene comparisons

Where designs are compared per gene g (full vs ablated model), differences d_g are ranked by magnitude and

$$W = \sum_{g: d_g > 0} \text{rank}(|d_g|), \quad p \text{ from the exact null distribution of } W, \quad (10)$$

a distribution-free paired test whose $W = 0$ (all differences one direction) is the strongest possible ordinal outcome at a given n .